

# HW 14

Due Wednesday, July 22, at 11:59pm.

## Starting a new Quarto file

For this assignment you will begin a new Quarto file. To do this, go to RStudio and click File > New File > Quarto Document. A pop-up will appear asking for a title, author, and format. You may leave these as the default options for now. Click OK and a new file will open. Update the YAML with your title, name, and date.

## Data

We will be using the `beijing` data set from HW 13.

Of interest are the following variables:

- `PM2.5`: PM2.5 levels in micrograms per cubic meter
- `TEMP`: temperature in degrees Celsius
- `PRES`: barometric pressure in hPa
- `wd`: wind direction (eight compass points)

You may load the dataset with the code below: (Ensure that you suppress any warnings and messages.)

```
beijing <- read.csv("data/beijing.csv", row.names = NULL)
```

## Exercise 1

- Fit a linear model that predicts PM2.5 levels based on the temperature, pressure, and wind direction. Write out the fitted model (in words and/or symbols). Do not include interactions. You may have to write your model on multiple lines for it to fit on the page – please check before submitting!
- What PM2.5 level would you predict for a day with temperature of 20 degrees C< pressure of 1040 hPa, and wind coming from the south?
- Provide and interpret the estimates for the slope corresponding to `PRES` and for the slope corresponding to `wdSw`.
- Provide and interpret your adjusted  $R^2$  value from this model.

## Exercise 2

- Fit a new linear model that predicts PM2.5 levels based on the temperature, barometric pressure, and their interaction. Write out the fitted model (in words and/or symbols).
- Based on your model in Exercise 2a, what would be the expected difference in PM2.5 levels give a one degree increase in temperature?
- Is there sufficient evidence that the relationship between temperature and PM2.5 depends on the barometric pressure? Briefly explain.

## AI Attestation

Was AI used on this assignment? If so, describe your prompts below. If not, simply state “AI was not used on this assignment.”

## Submission

As you’ve seen previously, we can **Render** into an .html file that can be opened by any web browser. To export it as a .pdf, open the file in your web browser and then print to or save as a .pdf document. Contact me in Ed Discussion if you need help!

You will submit the PDF documents for labs and homework to Gradescope as part of your final submission.

To submit your assignment:

- Access Gradescope through the menu on the BIOS 600 Canvas site.
- Click on the assignment, and you’ll be prompted to submit it.
- Mark the pages associated with each exercise. All of the pages of your lab should be associated with at least one question (i.e., should be “checked”).
- Select the first page of your .PDF submission to be associated with the “Formatting” section.

## Grading

| Component      | Points |
|----------------|--------|
| Ex 1           | 6      |
| Ex 2           | 5      |
| AI Attestation | 1      |
| Formatting     | 3      |

The “Formatting” grade is to assess the document format. This includes having a neatly organized document (no excessive output, warnings/messages when loading packages and/or data) with readable code and your name and the date updated in the YAML.